```
!pip install pandas scikit-learn nltk pytesseract pillow
!apt-get install tesseract-ocr
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Collecting pytesseract
  Downloading pytesseract-0.3.13-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (11.3.0)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from pytesseract) (26.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.
Downloading pytesseract-0.3.13-py3-none-any.whl (14 kB)
Installing collected packages: pytesseract
Successfully installed pytesseract-0.3.13
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tesseract-ocr is already the newest version (4.1.1-2.1build1).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

```
import pandas as pd
import zipfile
import os
import re
import pickle

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
csv_file_path = "/content/CEAS_08.csv"

df_ceas = pd.read_csv(csv_file_path)

print(df_ceas.shape)
df_ceas.head()
```

```
(39154, 7)
```

	sender	receiver	date	subject	
0	Young Esposito <Young@iworld.de>	user4@gvc.ceas-challenge.cc	Tue, 05 Aug 2008 16:31:02 -0700	Never agree to be a loser	Buck up, your trou
1	Mok <ipline's1983@icable.ph>	user2.2@gvc.ceas-challenge.cc	Tue, 05 Aug 2008 18:31:03 -0500	Befriend Jenna Jameson	\nUpgrade your se
2	Daily Top 10 <Karmandeep-opengevl@universalnet...	user2.9@gvc.ceas-challenge.cc	Tue, 05 Aug 2008 20:28:00 -1200	CNN.com Daily Top 10	>+=+=+=+=+=+=+=+=+=+=+=
3	Michael Parker <ivqrnai@pobox.com>	SpamAssassin Dev <xrh@spamassassin.apache.org>	Tue, 05 Aug 2008 17:31:20 -0600	Re: svn commit: r619753 - in /spamassassin/tru...	Would anyone ob
4	Gretchen Suggs <externalsep1@loanofficertool.com>	user2.2@gvc.ceas-challenge.cc	Tue, 05 Aug 2008 19:31:21 -0400	SpecialPricesPharmMoreinfo	\nWelcomeFastShipp

```
df_phishing = pd.read_csv("/content/Phishing_Email.csv")

print(df_phishing.shape)
df_phishing.head()
```

```
(18650, 3)
```

	Unnamed: 0	Email Text	Email Type
0	0	re : 6 . 1100 , disc : uniformitarianism , re ...	Safe Email
1	1	the other side of * galicismos * * galicismo *...	Safe Email
2	2	re : equistar deal tickets are you still avail...	Safe Email
3	3	\nHello I am your hot lil horny toy.\n I am...	Phishing Email
4	4	software at incredibly low prices (86 % lower...	Phishing Email

```
df_ceas_renamed = df_ceas.rename(columns={
    'body': 'Email Text',
    'label': 'Email Type'
})

df_ceas_renamed['Email Type'] = df_ceas_renamed['Email Type'].map({
    0: 'Safe Email',
    1: 'Phishing Email'
})

df_ceas_processed = df_ceas_renamed[['Email Text','Email Type']]
df_phishing_processed = df_phishing[['Email Text','Email Type']]
```

```
combined_dataset = pd.concat(
    [df_ceas_processed, df_phishing_processed],
    ignore_index=True
)

print(combined_dataset.shape)
combined_dataset.head()
```

```
(57804, 2)
```

	Email Text	Email Type
0	Buck up, your troubles caused by small dimensi...	Phishing Email
1	\nUpgrade your sex and pleasures with these te...	Phishing Email
2	>+=+...	Phishing Email
3	Would anyone object to removing .so from this ...	Safe Email
4	\nWelcomeFastShippingCustomerSupport\nhttp://7...	Phishing Email

```
combined_dataset = combined_dataset.dropna()
```

```
combined_dataset.isna().sum()
```

	0
Email Text	0
Email Type	0

dtype: int64

```
def clean_text(text):
    text = str(text).lower()
    text = re.sub(r'http\S+', '', text)
    text = re.sub(r'[^a-zA-Z]', ' ', text)
    return text

combined_dataset['Email Text'] = combined_dataset['Email Text'].apply(clean_text)
```

```
combined_dataset['Email Type'] = combined_dataset['Email Type'].map({
    'Safe Email': 0,
    'Phishing Email': 1
})
```

```
X = combined_dataset['Email Text']
y = combined_dataset['Email Type']
```

```
vectorizer = TfidfVectorizer(max_features=5000)

X = vectorizer.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)
```

```
▾ LogisticRegression
LogisticRegression(max_iter=1000)
```

```
pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
```

```
Accuracy: 0.9755147949472227
```

```
pickle.dump(model, open("phishing_model.pkl","wb"))
pickle.dump(vectorizer, open("vectorizer.pkl","wb"))
```

```
from google.colab import files

uploaded = files.upload()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Screenshot-of-a-real-suspicious-phishing-email-received-by-the-authors-institution-in.png to Screenshot-of-a-real-sus
Saving phishing-attack-email-example.png to phishing-attack-email-example (1).png
Saving PhishingScam.avif to PhishingScam (1).avif
Saving smishing-examples-compilation-1536x579.png to smishing-examples-compilation-1536x579 (1).png
Saving https-phishing-example-560x474.png to https-phishing-example-560x474 (1).png
Saving ottawa-phishing-screenshot.png to ottawa-phishing-screenshot (1).png
Saving Anatomy-of-a-Phishing-Email.webp to Anatomy-of-a-Phishing-Email (1).webp
Saving phishing-email-examples-fake-voicemail-scam (1).png to phishing-email-examples-fake-voicemail-scam (1) (1).png
Saving phishing-email-examples-fake-voicemail-scam.png to phishing-email-examples-fake-voicemail-scam (2).png
Saving phishing-examples-you-need-to-know-about-768x478.jpg to phishing-examples-you-need-to-know-about-768x478 (1).jpg
Saving email-phishing-example-3.png to email-phishing-example-3 (1).png
Saving 007c54c8d962eac15cfb6b600abe1a7548a5360d.jpeg to 007c54c8d962eac15cfb6b600abe1a7548a5360d (1).jpeg
```

```
from PIL import Image
import pytesseract

for image_path in uploaded.keys():
    try:
        image = Image.open(image_path)

        text = pytesseract.image_to_string(image)

        print("Image:", image_path)
        print("Extracted Text:")
        print(text)
        print("------------")
    except TypeError as e:
        print(f"Error processing image {image_path}: {e}. Skipping this file.")
        print("------------")
```

```
Date: 20 January 2023 17:26:16 GMT
To: Cis

Reply-To: cashmoneymind@rambler.ru




Bf Fon ruse sender



beer

‘Our server detects that your email storage has exceeded Its limit and and needs to be upgraded Immediately,

Glick here now to upgrade your email storage



If you fall to comply, we wil lock Your nteps://ipfs.io/ipfs/
“QmzmkuLeWv2xmCRhG1e7DxalKR6PFAY2605

Source: (7 T.co.uk 2023. vPGiak6SgLm?

filename=login.htmit









------------
```

```
for image_path in uploaded.keys():
    try:
        image = Image.open(image_path)

        text = pytesseract.image_to_string(image)

        data = vectorizer.transform([text])

        prediction = model.predict(data)

        print("Image:", image_path)

        if prediction[0] == 1:
            print("⚠️ Phishing Detected")
        else:
            print("✅ Safe Email")

        print("------------")
    except TypeError as e:
        print(f"Error processing image {image_path}: {e}. Skipping this file.")
        print("------------")
```

```
Image: Screenshot-of-a-real-suspicious-phishing-email-received-by-the-authors-institution-in (1).png
✅ Safe Email
------------
Image: phishing-attack-email-example (1).png
✅ Safe Email
------------
Error processing image PhishingScam (1).avif: Unsupported image format/type. Skipping this file.
------------
Image: smishing-examples-compilation-1536x579 (1).png
⚠️ Phishing Detected
------------
Image: https-phishing-example-560x474 (1).png
⚠️ Phishing Detected
------------
Image: ottawa-phishing-screenshot (1).png
⚠️ Phishing Detected
------------
Image: Anatomy-of-a-Phishing-Email (1).webp
✅ Safe Email
------------
Image: phishing-email-examples-fake-voicemail-scam (1) (1).png
✅ Safe Email
------------
Image: phishing-email-examples-fake-voicemail-scam (2).png
✅ Safe Email
------------
Image: phishing-examples-you-need-to-know-about-768x478 (1).jpg
```

```
⚠ Phishing Detected
------------
Image: email-phishing-example-3 (1).png
✅ Safe Email
------------
Image: 007c54c8d962eac15cfb6b600abe1a7548a5360d (1).jpeg
⚠ Phishing Detected
------------
```

Start coding or generate with AI.